

of online text! That's more text data than any other AI has ever learned from before. With this huge training dataset, GPT-3 gained an extremely advanced understanding of how human language works. It can generate sentences, have conversations, summarise long articles, translate languages, and more. GPT-3 seems to truly grasp the nuances and complexity of human writing and speaking. Let's see an example. Say we give GPT-3 this prompt: 'Write a short children's story about a lost dog trying to find his way home'. Here is one possible story GPT-3 could generate in response:

Rex the dog was wandering down the busy street, sad and alone. He had accidentally chased a car and gotten lost from his family! He missed playing fetch with the kids and cuddling on the couch at night.

Rex didn't know which way was home. He walked for a long time, asking people for help finding his house. But everyone was too busy to pay attention. Rex became hungry and tired. He napped on the sidewalk hoping his family would find him soon. Just as Rex was losing hope, he heard a familiar sound - the ice cream truck jingle! He remembered his home was always near where the ice cream truck parked. Rex followed the music all the way back to his street. His tail wagged wildly when he saw his house. The whole family shouted with joy and gave Rex a big hug when he came home. Rex was never lost again, and made sure to stay close when outside. But he knew if he got confused, he could listen for the ice cream truck to guide him. Rex was overjoyed to be back where he belonged.

Pretty impressive, right? The story has a clear beginning, middle and end. It has descriptive language, emotion, and fun kid-friendly details. GPT-3 learned all of this from studying so many

books and articles written by humans. While not perfect, its writing ability is incredibly advanced for AI!

The possibilities with GPT-3 are endless. It can answer tricky questions, summarise long reports into bullet points, suggest code fixes for programmers, and more. Some even believe it's a step towards an AI that can truly reason and think like humans. Hugging Face's Transformers library has played a crucial role in making advanced language models, including GPT-3, accessible to developers. It fosters collaboration and accelerates innovation in the AI community.

Applications of GPT-3

Tools like GPT-3 are generally used for content creation, coding assistance, language translation, and chatbots. There are a few other applications of GPT-3 too.

Speech synthesis: GPT-3 can generate synthetic speech from text that sounds more natural, and has better intonation and emphasis.

Machine translation: Fine-tuned versions of GPT-3 significantly improve the quality of language translations compared to previous models.

Answering questions: It provides relevant and coherent answers to queries, drawing conclusions from textual evidence. This is helpful for search, education, and research.

Summarisation: It can distil the key points from long reports, research papers, articles and other documents into concise summaries.

Creativity and idea generation: GPT-3 can expand on prompts to generate ideas for stories, articles, ad campaigns, product names and other creative works.

GPT-3's limitations

While impressive in certain applications, GPT-3 has several shortcomings that are important to recognise. At first glance, GPT-3 seems amazingly capable - the hype makes it sound like a human-level AI. But under closer scrutiny, fundamental

weaknesses become apparent. This is because it is only as good as the data it was trained on.

Accuracy issues: GPT-3 can produce answers that sound plausible but may not always be accurate. It doesn't have the ability to fact-check information, so its responses should be verified for correctness.

Inability to generate personal experiences: GPT-3 doesn't have personal experiences or emotions. If you ask it about its own feelings or experiences, it may generate responses based on general knowledge rather than personal understanding.

Redundancy: In certain situations, GPT-3 may provide more information than necessary or generate verbose responses. This can lead to outputs that sound plausible but are longer than needed.

No real-time learning: GPT-3 does not learn or adapt in real-time. Once it's trained, it remains static until the next training update. It doesn't learn from new information or experiences after its training data cut-off.

Building GPT-3 with Python

In the October 2023 issue of *Open Source For You* (page 72), I discussed how we can create our own ChatGPT bot in Telegram using an OpenAI API token. For your convenience I have attached my article link in the QR code box in Figure 1. Do go through that. We will now discuss building GPT-3 in the terminal. So, fire up your PC and start



Figure 1: Link to creating a ChatGPT bot in Telegram